



Federal Board HSSC-I Examination

Mathematics Model Question Paper

National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)
(50% MCQs & 50% Descriptive)

SECTION - A (Marks 40)

Time Allowed: 1 hour 40 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

ROLL NUMBER					
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Version No.			
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Candidate Sign. _____

Invigilator Sign. _____

Note: Fill the correct bubble for each question. Each question carries 0.75 mark.

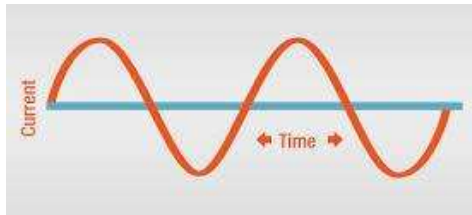
S No.	Question	A	B	C	D	A	B	C	D
1.	The linear factors of the polynomial $4z^2 + 9y^2$ are:	$(2z - 3yi)^2$	$(2z + 3yi)^2$	$(2z + 3y)(2z - 3y)$	$(2z + 3yi)(2 - 3yi)$	☐	☐	☐	☐
2.	What is the principal argument of $-i - \sqrt{3}$?	$-\frac{4\pi}{3}$	$\frac{4\pi}{3}$	$-\frac{\pi}{3}$	$\frac{\pi}{3}$	☐	☐	☐	☐
3.	If $z = r\cos\theta + ir\sin\theta$ then z^{-1} is equal to:	$\frac{\cos\theta - isin\theta}{r}$	$\frac{-\cos\theta + isin\theta}{r}$	$\frac{-\cos\theta - isin\theta}{r}$	$\frac{\cos\theta + isin\theta}{r}$	☐	☐	☐	☐
4.	If A is a square matrix of order 3 with $ A = 5$, then the value of $ 2A $ is:	5	10	25	40	☐	☐	☐	☐
5.	For $A = \begin{bmatrix} 3 & 3 & 1 \\ 0 & 5 & -1 \\ 2 & 1 & 4 \end{bmatrix}$, what is the value the cofactor A_{12} ?	-2	0	2	-11	☐	☐	☐	☐
6.	Which condition defines an inconsistent system of linear equations?	One solution	Infinite solutions	Trivial solution	No solution	☐	☐	☐	☐
7.	If the first term of a sequence is 10 and the common difference is 0, what is the 50th term?	0	10	50	500	☐	☐	☐	☐
8.	Row 1 of a theatre has 20 seats. Each successive row has 2 additional seats. At which row number does the seat count reach 40?	10th	11th	12th	20th	☐	☐	☐	☐
9.	A ball is dropped and rebounds to 60%, of its previous height. If the first bounce is 100 cm, what is the height of the 2 nd bounce?	60 cm	48 cm	36 cm	6 cm	☐	☐	☐	☐

10.	Find the 5 th term of an HP whose corresponding AP is 5, 10, 15, ...	$\frac{1}{20}$	$\frac{1}{25}$	$\frac{1}{30}$	$\frac{1}{35}$	☐ ☐ ☐ ☐
11.	The third term in the expansion of $(x^2-2)^4$ is:	$24x^4$	$-24x^4$	$-32x^2$	$32x^2$	☐ ☐ ☐ ☐
12.	Which term in the expansion of $(x^2 - \frac{1}{x^2})^6$ is independent of x ?	T_3	T_4	T_5	T_6	☐ ☐ ☐ ☐
13.	The simplified form of $(x+1)^3 - (x-1)^3$ is:	$2x^3 + 6x$	$6x^2 + 2$	$6x^2 + 2x$	$2x^3 + 6x^2$	☐ ☐ ☐ ☐
14.	When 3^{50} is divided by 8, the remainder is:	1	3	5	7	☐ ☐ ☐ ☐
15.	When dividing $x^3 + 2x^2 - 5x + 1$ by $x - 2$, the degree of the remainder is:	0	1	2	7	☐ ☐ ☐ ☐
16.	Dividing $x^4 - 3x^2 + 2$ by $x^2 - 1$, the quotient is:	$x^2 - 2$	$x^2 + 2$	$x^2 - 1$	$x^2 + 1$	☐ ☐ ☐ ☐
17.	For what value of k , $x - 2$ is the factor of $x^3 - kx^2 + 4x - 4$?	-3	0	3	4	☐ ☐ ☐ ☐
18.	Two direction angles of vector $\vec{a}$ are 30° and 60° . What is the value of third direction angle?	135°	90°	45°	30°	☐ ☐ ☐ ☐
19.	The vector parallel to $2\hat{i} + 6\hat{j} - 5\hat{k}$ and with magnitude 3 is:	$[\frac{12}{\sqrt{65}}, \frac{-18}{\sqrt{65}}, \frac{75}{\sqrt{65}}]$	$[\frac{12}{\sqrt{65}}, \frac{18}{\sqrt{65}}, \frac{25}{\sqrt{65}}]$	$[\frac{6}{\sqrt{65}}, \frac{18}{\sqrt{65}}, \frac{-15}{\sqrt{65}}]$	$[\frac{6}{\sqrt{65}}, \frac{-18}{\sqrt{65}}, \frac{8}{\sqrt{65}}]$	☐ ☐ ☐ ☐
20.	The area of a parallelogram having adjacent sides $\vec{a} = [2, 5, -2]$ and $\vec{b} = [1, 4, -2]$ is:	$\sqrt{17}$	$\frac{\sqrt{17}}{2}$	$\frac{\sqrt{17}}{6}$	$2\sqrt{17}$	☐ ☐ ☐ ☐
21.	At which angle the work done is zero?	0°	45°	90°	180°	☐ ☐ ☐ ☐
22.	In navigation the velocity of the air-plane is calculated using:	Dot product	Vector addition	Vector subtraction	Cross product	☐ ☐ ☐ ☐
23.	If $\vec{a} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} - 3\hat{k}$ then $\hat{i} \cdot \vec{a} + \hat{j} \cdot \vec{b} = ?$	1	-2	2	4	☐ ☐ ☐ ☐
24.	If $\sin \theta = \frac{3}{5}$ and θ is acute, what is the value of $\sin 2\theta$?	$\frac{24}{25}$	$\frac{12}{25}$	$\frac{7}{25}$	$\frac{16}{25}$	☐ ☐ ☐ ☐
25.	If $\cos \theta = -\frac{1}{3}$ and θ lies in quadrant II, the value of $\sin(\frac{\theta}{2})$ is:	$\frac{\sqrt{6}}{3}$	$-\frac{\sqrt{6}}{3}$	$\frac{\sqrt{3}}{3}$	$-\frac{\sqrt{3}}{3}$	☐ ☐ ☐ ☐
26.	Simplifying $2(\sin 3x)(\cos x)$ into a sum or difference we get:	$\sin 4x + \sin 2x$	$\sin 4x - \sin 2x$	$\sin 2x - \sin x$	$\sin 2x + \sin x$	☐ ☐ ☐ ☐
27.	If $\tan \alpha = \frac{1}{2}$ and $\tan \beta = \frac{1}{3}$, (α, β are acute), the exact value of $(\alpha + \beta)$ is:	0°	8°	45°	90°	☐ ☐ ☐ ☐
28.	Which of the following is equivalent to $\sin(270^\circ - \theta)$?	$-\sin \theta$	$-\cos \theta$	$\sin \theta$	$\cos \theta$	☐ ☐ ☐ ☐
29.	If $\sin 25^\circ = t$, then $\cos 50^\circ$ in terms of t is:	$2t^2 - 1$	$2t\sqrt{1-t^2}$	$1 - 2t^2$	$\sqrt{1-2t^2}$	☐ ☐ ☐ ☐

30.	What is the range of the function $y = 3\sin(2x)$?	$[-1, 1]$	$[-2, 2]$	$[-3, 3]$	$(-\infty, \infty)$	☐ ☐ ☐ ☐
31.	The sum of $\sin(-x) + \sin x$ is?	0	1	$2\sin x$	$\sin 2x$	☐ ☐ ☐ ☐
32.	The period of $4\cos\left(\frac{x}{3}\right)$ is:	$\frac{4\pi}{3}$	$\frac{2\pi}{3}$	2π	6π	☐ ☐ ☐ ☐
33.	What is the maximum value of $10\sin x - 5\cos x + 2$?	3	17	7	13	☐ ☐ ☐ ☐
34.	Which pair of the trigonometric functions has period π ?	$\sin x, \cos x$	$\tan x, \cot x$	$\sec x, \csc x$	$\sin x, \tan x$	☐ ☐ ☐ ☐
35.	A pizza shop offers 8 toppings. How many different pizzas can be made if you must choose exactly 3 toppings?	24	56	92	336	☐ ☐ ☐ ☐
36.	In how many ways can a student choose 4 questions out of 6 in an exam?	15	24	30	360	☐ ☐ ☐ ☐
37.	In how many ways a DJ can play 5 songs out of 12 (order matters)?	$\binom{12}{5}$	$\frac{12!}{5!}$	${}^{12}P_5$	$\frac{12!}{4!}$	☐ ☐ ☐ ☐
38.	If a coin is tossed 3 times, what is the total number of possible outcomes?	3	6	8	9	☐ ☐ ☐ ☐
39.	The value of ${}^nP_{n-1}$ is?	n	$n!$	$\frac{n!}{(2n-1)!}$	$\frac{n}{2n-1}$	☐ ☐ ☐ ☐
40.	If ${}^{n-1}P_3 : {}^nP_4 = 1:9$ then value of n is:	3	6	7	9	☐ ☐ ☐ ☐

STIMULUS/SITUATION/SCENARIO BASE MCQs

Scenario for MCQs (41-44)		Given a complex polynomial $z^2 + 4$, where z is a complex number.				
41.	The complex polynomial $z^2 + 4$ is:	reducible to real factors	irreducible to real factors	reducible to complex factors	irreducible to complex factors	☐ ☐ ☐ ☐
42.	If 4 is replaced with 8 and z^2 is replaced with z^3 in polynomial $(z^2 + 4)$, then the number of linear factor(s) of polynomial are:	1	2	3	Not possible	☐ ☐ ☐ ☐
43.	The linear factors of polynomial $z^2 + 4$ are:	$z + 4i,$ $z - 4i$	$z - 2i,$ $z - 2i$	$z + 2i,$ $z - 4i$	$z + 2i,$ $z - 2i$	☐ ☐ ☐ ☐
44.	The two factors of $z^2 + 4$ are:	conjugate of each other	same	additive inverse of each other	reciprocal of each other	☐ ☐ ☐ ☐
Scenario for MCQs (45-48)		Three seismic stations track wave data where x, y , and z represent the exact amplitude of primary, secondary, and surface waves respectively. The seismic network model is: $AX = B$, $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 5 & 5 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 0 \end{bmatrix}$, then				
45.	The value of $ A $ is:	-16	-8	0	16	☐ ☐ ☐ ☐
46.	The value of $ A_z $ is:	-16	-60	0	60	☐ ☐ ☐ ☐
47.	The value of z is:	-3.75	0.27	3.75	7.5	☐ ☐ ☐ ☐

48.	The value of $ A $ if Row 3 is replaced by $\begin{bmatrix} 3 & 3 & 3 \end{bmatrix}$	-48	-16	0	48	☐ ☐ ☐ ☐										
	Scenario for MCQs (49-52)	Use the table below showing the Weekly Savings of a student to answer the questions.														
		<table><tr><th>Week (n)</th><th>Total Savings (S_n)</th></tr><tr><td>1st</td><td>Rs 150</td></tr><tr><td>2nd</td><td>Rs 350</td></tr><tr><td>3rd</td><td>Rs 600</td></tr><tr><td>4th</td><td>Rs 900</td></tr></table>		Week (n)	Total Savings (S_n)	1 st	Rs 150	2 nd	Rs 350	3 rd	Rs 600	4 th	Rs 900			
Week (n)		Total Savings (S_n)														
1 st		Rs 150														
2 nd		Rs 350														
3 rd	Rs 600															
4 th	Rs 900															
49.	What is the value of a_2 ?	150	200	250	350	☐ ☐ ☐ ☐										
50.	What is the total savings for the 3rd week?	200	250	350	600	☐ ☐ ☐ ☐										
51.	Which formula represents the total savings (S_n) for any week (n)?	$S_n = \frac{n}{2}[300 + (n-1)50]$	$S_n = 150 + (n-1)50$	$S_n = \frac{n}{2}[150 + (n-1)50]$	$s_n = 150n + 50$	☐ ☐ ☐ ☐										
52.	By the end of which week will the student's total savings first exceed Rs. 2,500?	6th week	7th week	8th week	9th week	☐ ☐ ☐ ☐										
	Scenario for MCQs (53-56)	A drone is flying in space. Its motion is represented by the vectors $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. An engineer wants to analyze the relationship between these two directions using dot and cross products.														
53.		The engineer computes $\vec{a} \cdot \vec{b}$. What does this quantity help to determine?	The area of parallelogram formed by the vectors	The angle between the vectors	Law of sines	The area of a triangle	☐ ☐ ☐ ☐									
54.		In terms of vectors $\vec{a}$ and $\vec{b}$, how would you express the work done on the drone?	$ \vec{a} + \vec{b} $	$ \vec{a} - \vec{b} $	$\vec{a} \times \vec{b}$	$\vec{a} \cdot \vec{b}$	☐ ☐ ☐ ☐									
55.		If $\vec{a} \times \vec{b} = 0$. What can be concluded about the drone's angle of direction?	0°	45°	60°	90°	☐ ☐ ☐ ☐									
56.		An engineer wants to position a communication antenna perpendicular to both $\vec{a}$ and $\vec{b}$. Which vector gives its orientation?	$-3\hat{i} + 3\hat{k}$	$3\hat{i} + 3\hat{k}$	$-3\hat{i} - 3\hat{k}$	$3\hat{i} - 3\hat{k}$	☐ ☐ ☐ ☐									
	Scenario for MCQs (57-60)	Current in the circuit with the passage of time follows the wave pattern.														
																
57.		Wave shown in the figure is often known as:	secant wave	tangent wave	sine wave	longitudinal wave	☐ ☐ ☐ ☐									
58.		If the function $y = 2\sin x$ represents the wave, then what is the domain of the function?	$(-\infty, 0]$	$[0, \infty)$	$(-\infty, \infty)$	$[1, \infty)$	☐ ☐ ☐ ☐									
59.		What is the maximum value of $y = 2\sin x$?	0	1	2	∞	☐ ☐ ☐ ☐									
60.	If $y = 2\sin x$ is replaced by $y = 3\cos x$, then what is the range of the new function?	$(-3, 3)$	$[-3, 3]$	$(-3, 3]$	$[-3, 3)$	☐ ☐ ☐ ☐										
	Scenario for MCQs (61-64)	A student council has 12 members: 5 from Grade 11 and 7 from Grade 12. A committee of 4 members is to be formed. The committee must have exactly 2 students from Grade 11 and 2 from Grade 12. The order of selection does not matter.														

61.	The number of ways to choose 2 students from Grade 11 is:	5	10	20	25	☐ ☐ ☐ ☐
62.	How many ways to choose 2 students from Grade 12?	7	14	21	42	☐ ☐ ☐ ☐
63.	The total number of ways to form the committee is:	31	210	420	840	☐ ☐ ☐ ☐
64.	If the committee instead had no grade restriction, the number of possible committees would be:	210	330	495	792	☐ ☐ ☐ ☐



Federal Board HSSC-I Examination
Model Question Paper Mathematics
National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)
(50% MCQs & 50% Descriptive)

Time allowed: 1.20 hours

Total Marks: 52

Note: Answer all parts from Section 'B' and all questions from Section 'C' on the **E-sheet**.
Write your answers on the allotted/given spaces.

SECTION – B (Marks 52)

(7 × 4 = 28)

Q.No	Question	Marks	Question	Marks
1.	Find the quadratic equation of: $\arg(z - 2) - \arg(z + 2) = \pi/4$	4	OR Find A^{-1} , If $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ 3 & 7 & 7 \end{bmatrix}$ by using row operations.	4
2.	The 2 nd term of an H.P. is $\frac{1}{5}$ and 5 th term is $\frac{1}{11}$. Find the 12 th term?	4	OR Solve the simultaneous linear equations: $\frac{3}{2}z - 6\left(1 + \frac{i}{3}\right)w = 5,$ $\frac{i}{2}z + \left(\frac{3}{4} - \frac{1}{2}i\right)w = \left(\frac{1}{2} + 2i\right)$	4
3.	If the vectors $\vec{u} = p\hat{i} + \hat{j} + \hat{k}$, $\vec{v} = \hat{i} + q\hat{j} + \hat{k}$ and $\vec{w} = \hat{i} + \hat{j} + r\hat{k}$ are coplanar, then prove that $\frac{1}{1-p} + \frac{1}{1-q} + \frac{1}{1-r} = 1$	4	OR Divide $3x^4 - 2x^3 + 4x^2 - x + 5$ by $x^2 - x + 1$. Find the quotient and remainder.	4
4.	In a class of 40 students, 25 passed Mathematics and 18 passed Science. A student claims: "At least 3 students passed both subjects." Using the principle of inclusion-exclusion, justify whether this claim is correct.	4	OR Express the vector $\vec{a} = 3\hat{i} + 7\hat{j} + 2\hat{k}$ as a sum of two vectors one of which is parallel and other is perpendicular to the vector $\vec{b} = 5\hat{i} + \hat{j} + 8\hat{k}$.	4
5.	Given: $y = 2 \cos(5x) \cos(x) + 2 \sin(3x) \sin(x)$ i. Simplify y into a single sum or difference of cosine functions. ii. Calculate the exact numerical value of y when $x = 15^\circ$.	4	OR In an A.P, the sum of the first n terms is given by: $S_n = 3n + tn^2$. If $S_4 = 28$, i. Find the value of t . ii. Determine first term and the common difference.	4
6.	In a lottery, a player selects 5 distinct numbers from 1 to 50. i. Calculate the total number of possible combinations. ii. If a player buys 5 tickets with different combinations, what is the probability of winning the jackpot (matching all 5 numbers)?	4	OR A Ferris wheel has a radius of 10 meters and its centre is 12 meters above the ground. It completes one revolution in 40 seconds. A passenger starts at the lowest point at time $t = 0$. i. Model the passenger's height $h(t)$ above the ground using a cosine function of the form $h(t) = a \cos(kt) + c$. Find the	4

				values of a , k , and c . ii. Calculate the passenger's height after 10 seconds.	
7.	Determine whether the function $f(x) = x \sin x$ is even, odd, or neither. Also, check whether it is periodic; if yes, find its fundamental period.	4	OR	Find n if $\frac{1}{2(n-2)!} : \frac{1}{4!(n-4)!} = 2:1$ for $n \geq 4$	4

SECTION – C (Marks 24)

Note: Attempt all questions. Marks of each question are given. ($3 \times 8 = 24$)

Q. No.	Question	Marks		Question	Marks
Q8	Consider the system of linear equations: $\begin{cases} 3x + y - \beta z = 0 \\ 4x - 2y - 3z = 0 \\ 2x + 4y + \beta z = 0 \end{cases}$ Find the value(s) of β for which the given system of homogeneous equations has non-trivial solution. Also solve the system for the value(s) of β .	8	OR	Expand $(3 + x)^5$ using the Binomial Theorem. Then evaluate the expansion at $x = 0.1$ to find the numerical value of $(3.1)^5$ without a calculator (show terms).	8
Q9	Given that $1 + \frac{4}{5} + \frac{7}{5^2} + \frac{10}{5^3} + \dots$ i. Find the sum of first n terms of the series. ii. Find the sum to infinity of the series.	8	OR	iii. Given the cubic polynomial iv. $Q(x) = x^3 - 6x^2 + 11x - 6$. v. Use the Factor Theorem to find one linear factor by testing integral roots. vi. Perform polynomial division to find the quadratic factor, then factorize completely.	8
Q10	If $\vec{a} = 3\hat{i} - \hat{j} + 5\hat{k}$, $\vec{b} = 4\hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} + 5\hat{j} + \hat{k}$, then verify that $\vec{a} \cdot \vec{b} \times \vec{c} = \vec{b} \cdot \vec{c} \times \vec{a} = \vec{c} \cdot \vec{a} \times \vec{b}$.	8	OR	Show that $\cos 5\theta + 2\cos 3\theta + \cos \theta = 4 \cos 3\theta \cos^2 \theta$ and verify the identity for $\theta = 20^\circ$	8